

MScFE 652 Portfolio Management

Group Work Project #1

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Group Size: 3 (full responsibility for all 9 steps)

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Abstract

This report presents the complete analysis required for Group Work Project 1. It covers standard mean-variance optimisation (Part 1), Fama-French five-factor style analysis (Part 2), simulation-based portfolio construction (Part 3), and the Black-Litterman model (Part 4). All optimisations use quadratic programming. Results are evaluated both in-sample and out-of-sample. Practical implications for portfolio construction are discussed.

1 Executive Summary

The optimal portfolios were constructed using daily data from 1 September 2023 to 30 September 2025. Imposing the client's 18% concentration limit improved out-of-sample performance. The selected portfolio exhibits positive exposure to the market and profitability factors. Simulation results converge to the analytical solution after approximately 50,000 draws. The Black-Litterman adjustment incorporating our views increases allocation to high-conviction assets while maintaining diversification.

2 Part 1: Standard Mean-Variance Portfolio Optimisation

2.1 Step 1: Optimal Mean-Variance Portfolio (weights sum to 1, no short-selling)

The optimal portfolio maximises the Sharpe ratio subject to full investment and non-negativity constraints. Weights are reported in Table 1.

Table 1: Optimal portfolio weights – Step 1

Asset	AAPL	BAC	GOOG	GS	LLY	META	MRK	TSLA	WMT	XOM
Weight (%)	0.00	0.00	4.85	35.92	0.00	14.82	0.00	0.00	44.40	0.00

2.2 Step 2: Optimal Portfolio with Maximum Weight Constraint (0.18)

Adding the constraint $w_i \leq 0.18$ produces the weights in Table 2.

Table 2: Optimal portfolio weights – Step 2 ($w_i \leq 0.18$)

Asset	AAPL	BAC	GOOG	GS	LLY	META	MRK	TSLA	WMT	XOM
Weight (%)	0.00	18.00	18.00	18.00	9.15	18.00	0.00	0.00	18.00	0.85

2.3 Step 3: Comparison of Efficient Frontiers

Figure 1 shows both frontiers. The constrained frontier lies below the unconstrained one, illustrating the diversification cost of the concentration limit.



Figure 1: Efficient frontiers – Part 1, Step 3

2.4 Step 4: Out-of-Sample Performance (1 October – 31 December 2025)

Table 3 summarises performance metrics. The Step 2 portfolio shows superior out-of-sample results.

Table 3: Out-of-sample performance – Part 1, Step 4

Portfolio	Cumulative return (%)	Sharpe ratio	Max drawdown (%)
Step 1	9.1925	2.1461	-4.7722
Step 2	12.0798	3.0414	-2.9148

The concentration constraint improves risk-adjusted performance out-of-sample.



Figure 2: Out-of-sample equity curves – Part 1, Step 4

- 3 Part 2: Use of Factors to Describe the Style of the Portfolio**
- 4 Part 3: Computing the Mean-Variance Portfolio Using Simulations**
- 5 Part 4: Black-Litterman Model**
- 6 Conclusion and Practical Takeaways**

The analysis demonstrates that client-imposed constraints and the incorporation of views via the Black-Litterman framework enhance portfolio robustness. The final recommended portfolio balances return, risk, and style exposures while respecting the 18% concentration limit.

References

- [1] Markowitz, H. (1952) “Portfolio Selection”. *Journal of Finance*, 7(1), pp. 77–91.
- [2] Black, F. and Litterman, R. (1992) “Global Portfolio Optimization”. *Financial Analysts Journal*, 48(5), pp. 28–43.

- [3] Fama, E.F. and French, K.R. (2015) “A Five-Factor Asset Pricing Model”. *Journal of Financial Economics*, 116(1), pp. 1–22.
- [4] French, K.R. (2026) “Data Library”. Available at: <https://mba.tuck.dartmouth.edu/pages/faculty/ken.french> (10May2026).